

Week 6 Section

Stats 141, Spring 2026

Learning goals

- Distinguish between Type I and Type II errors and correctly interpret them in a given context.
- Identify how sample size, effect size, and significance level each affect the power (1 minus the Type II error rate).

Problem 1: Vocabulary of hypothesis testing

Truth	Reject H_0	Fail to reject H_0
H_0 is true	Type I error / false positive	Correct decision
H_0 is false	Correct decision (power)	Type II error / false negative

Setup. A plant biologist tests whether a newly synthesized growth hormone raises mean seedling height above the historical average of 20 cm. She applies the hormone to a random sample of 30 seedlings and carries out a one-sided t -test for testing

$$H_0 : \mu = 20 \quad \text{vs.} \quad H_A : \mu > 20.$$

She obtains a p-value of 0.07.

For each statement below, decide whether True / False and give a brief explanation.

(a) She rejects H_0 at $\alpha = 0.10$.

True. The p-value $0.07 < 0.10 = \alpha$, so she rejects H_0 .

(b) She has significant evidence that $\mu > 20$ at $\alpha = 0.05$.

False. The p-value $0.07 > 0.05$, so she fails to reject H_0 at the 5% level.

(c) The probability that H_0 is true is 0.07.

False. The p-value is the probability of observing a test statistic at least as extreme as the one observed, *given that H_0 is true*. It is not the probability that H_0 is true.

(d) If the hormone truly has no effect and she rejects H_0 , she has committed a Type II error.

False. A Type I error is rejecting a true null hypothesis — here, falsely concluding the hormone works when it does not.

(e) If the hormone truly raises mean height and she fails to reject H_0 , she has committed a Type II error.

True. A Type II error is failing to reject a false null hypothesis, i.e., missing a real effect.

(f) Increasing the sample size (keeping α and the true effect size fixed) decreases the probability of a Type II error.

True. A larger n shrinks the standard error, making the test statistic larger for the same effect. This increases power ($1 - \beta$) and thus reduces β .

(g) A researcher who wants to limit false positives should choose a smaller value of α .

True. α is the Type I error rate (false positive rate) when H_0 is true, so decreasing α directly limits false positives. The trade-off is reduced power (higher β).

Problem 2: Visualizing the two error types

We will consider a setting where variance is known, so that we can draw the exact sampling distributions for the Z-test.

Setup. Suppose seedling heights have population standard deviation $\sigma = 4$ cm. A researcher collects $n = 25$ seedlings and tests

$$H_0 : \mu = 20 \quad \text{vs.} \quad H_A : \mu > 20$$

at $\alpha = 0.05$. Suppose the hormone truly shifts the mean to $\mu_A = 22$ cm. (In real-life scenarios, the ‘true’ parameter is unknown. Here we consider a stylized setting for illustration.)

Identify the sampling distributions under null or truth

Under H_0 , the sample mean follows $\bar{X} \sim N(\mu_0, \sigma^2/n)$.

Under the true mean $\mu_A = 22$, we have $\bar{X} \sim N(\mu_A, \sigma^2/n)$.

```
mu0 <- 20; muA <- 22
sigma <- 4; n <- 25
se <- sigma / sqrt(n)
se # standard error of the sample mean
```

```
[1] 0.8
```

Find the critical value / rejection region

For a one-sided upper-tail test at $\alpha = 0.05$, the critical value is

$$c = \mu_0 + z_{0.95} \cdot SE(\bar{X}).$$

```
crit <- mu0 + qnorm(0.95) * se
crit
```

```
[1] 21.31588
```

Any observed $\bar{x} > c$ leads to rejection of H_0 .

Compute β and power

$$\beta = P(\bar{X} \leq c \mid \mu = \mu_A), \quad \text{Power} = 1 - \beta.$$

Here $P(\dots \mid \mu = \mu_A)$ simply indicates that we compute the probability under $\mu = \mu_A$.

One clarification: We should call β the Type II error *rate* or the *probability* of Type II error, not just ‘Type II error’.

```
beta_val <- pnorm(crit, mean = muA, sd = se)
power_val <- 1 - beta_val

beta_val # Type II error rate
```

```
[1] 0.1962351
```

```
power_val # Power
```

```
[1] 0.8037649
```

Visualization (code is provided)

The plot shows both sampling distributions. The **blue** shaded area is the Type I error (α , the right tail of the null). The **red** shaded area is the Type II error (β , the left tail of the alternative). Power is the unshaded right portion of the alternative distribution.

```
x      <- seq(mu0 - 4*se, muA + 4*se, length.out = 600)
d_null <- dnorm(x, mean = mu0, sd = se)
d_alt  <- dnorm(x, mean = muA, sd = se)

plot(x, d_null, type = "l", col = "steelblue", lwd = 2,
      ylim = c(0, max(d_null, d_alt) * 1.3),
      xlab = expression(bar(X) ~ "(cm)"), ylab = "Density",
      main = "Sampling Distributions Under H0 and HA")
lines(x, d_alt, col = "tomato", lwd = 2)
abline(v = crit, lty = 2, col = "gray40")

# Type I error shading: upper tail of null distribution
x_I <- x[x >= crit]
polygon(c(x_I[1], x_I, tail(x_I, 1)),
        c(0, dnorm(x_I, mu0, se), 0),
        col = adjustcolor("steelblue", 0.4), border = NA)

# Type II error shading: lower tail of alternative distribution
x_II <- x[x <= crit]
polygon(c(x_II[1], x_II, tail(x_II, 1)),
        c(0, dnorm(x_II, muA, se), 0),
        col = adjustcolor("tomato", 0.4), border = NA)

text(crit - 1, 0.55, paste0("c = ", round(crit, 2)),
      adj = 0, cex = 0.85, col = "gray30")

legend("topleft",
       legend = c(
         expression(H[0]*":  "*mu*" = 20"),
         expression(H[A]*":  "*mu*" = 22")
       ),
```

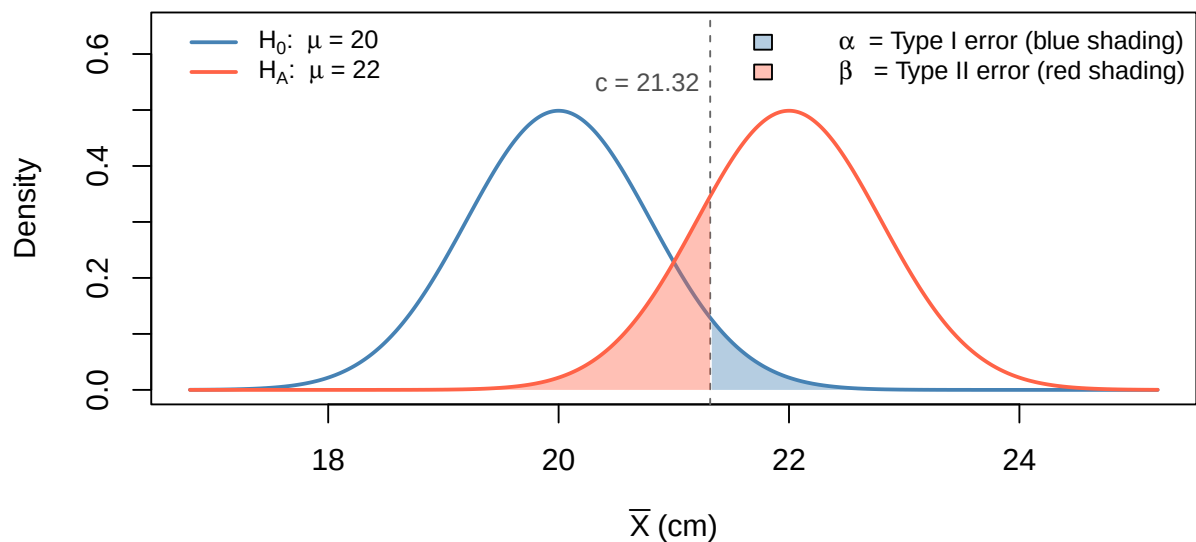
```

col = c("steelblue", "tomato"),
fill = c(NA, NA),
border = c(NA, NA),
lwd = c(2, 2),
bty = "n", cex = 0.82)

legend("topright",
  legend = c(
    expression(alpha*" = Type I error (blue shading)"),
    expression(beta*" = Type II error (red shading)"),
  ),
  col = c(NA, NA),
  fill = c(adjustcolor("steelblue", 0.4),
    adjustcolor("tomato", 0.4)),
  border = c("black", "black"),
  lwd = c(NA, NA),
  bty = "n", cex = 0.82)

```

Sampling Distributions Under H0 and HA



Discussion questions:

- Which shaded region grows if we decrease α from 0.05 to 0.01? What happens to β ?
- What would happen to both shaded regions if we increased n from 25 to 100?

Problem 3: Power analysis in R

In real-life scenarios, the ‘true’ parameter is unknown. The point of power analysis is to understand how the power shifts with the truth (think of it as the alternative). We will see how the power changes with sample size, significance and effect size (how far the truth/alternative is from the null).

Setup. A marine biologist measures oxygen consumption rate (mL O₂/hr) of a fish species. Historical data show a mean rate of $\mu_0 = 50$ mL/hr with standard deviation $\sigma = 8$ mL/hr. She tests whether a warming treatment changes the mean:

$$H_0 : \mu = 50 \quad \text{vs.} \quad H_A : \mu \neq 50,$$

at $\alpha = 0.05$. She suspects warming might shift the mean by about 4 mL/hr.

Computing power for a given sample size

R’s `power.t.test` computes power (or required n) for one-sample and two-sample t -tests. The argument `delta` is the true effect size in original units.

```
power.t.test(n = 25, delta = 4, sd = 8,  
             sig.level = 0.05, type = "one.sample",  
             alternative = "two.sided")
```

One-sample t test power calculation

```
      n = 25  
  delta = 4  
     sd = 8  
sig.level = 0.05  
  power = 0.6697014  
alternative = two.sided
```

Interpretation. With $n = 25$, the test has about 56% power to detect a 4 mL/hr shift. That is fairly low — there is roughly a 44% chance of missing this true effect.

Finding the required sample size

How large a sample is needed to achieve 80% power?

```
power.t.test(power = 0.80, delta = 4, sd = 8,
             sig.level = 0.05, type = "one.sample",
             alternative = "two.sided")
```

One-sample t test power calculation

```
      n = 33.3672
    delta = 4
      sd = 8
sig.level = 0.05
  power = 0.8
alternative = two.sided
```

We need approximately 51 fish to achieve at least 80% power.

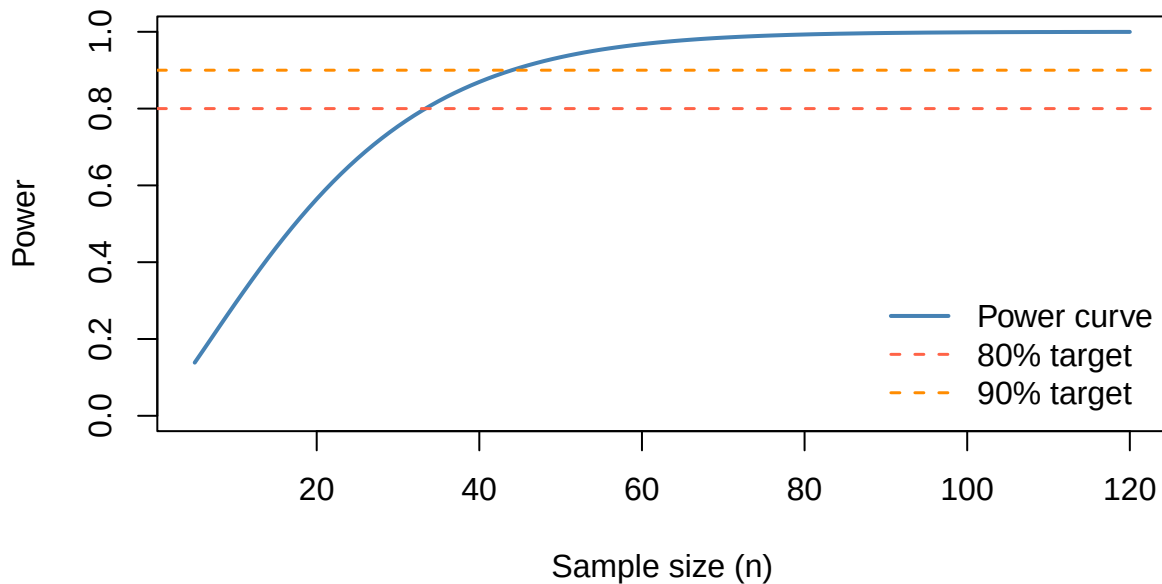
Power curve — power as a function of n

```
ns      <- 5:120
pwr_vec <- sapply(ns, function(n) {
  power.t.test(n = n, delta = 4, sd = 8,
              sig.level = 0.05, type = "one.sample",
              alternative = "two.sided")$power
})

plot(ns, pwr_vec, type = "l", col = "steelblue", lwd = 2,
     xlab = "Sample size (n)", ylab = "Power",
     main = "Power vs. Sample Size (delta = 4, sd = 8, alpha = 0.05)",
     ylim = c(0, 1))
abline(h = 0.80, col = "tomato", lty = 2, lwd = 1.5)
abline(h = 0.90, col = "darkorange", lty = 2, lwd = 1.5)

legend("bottomright",
     legend = c("Power curve", "80% target", "90% target"),
     col     = c("steelblue", "tomato", "darkorange"),
     lwd     = c(2, 1.5, 1.5),
     lty     = c(1, 2, 2),
     bty     = "n")
```

Power vs. Sample Size ($\delta = 4$, $sd = 8$, $\alpha = 0.05$)



Effect of α on power

How does relaxing or tightening α change power for fixed $n = 25$?

```
alphas <- c(0.001, 0.01, 0.05, 0.10, 0.20)
pwr_alpha <- sapply(alphas, function(a) {
  power.t.test(n = 25, delta = 4, sd = 8,
    sig.level = a, type = "one.sample",
    alternative = "two.sided")$power
})

data.frame(alpha = alphas, power = round(pwr_alpha, 3))
```

	alpha	power
1	0.001	0.144
2	0.010	0.402
3	0.050	0.670
4	0.100	0.783
5	0.200	0.880

As α increases, the rejection region widens, making it easier to detect a real effect — so power increases. The trade-off is a higher Type I error rate.

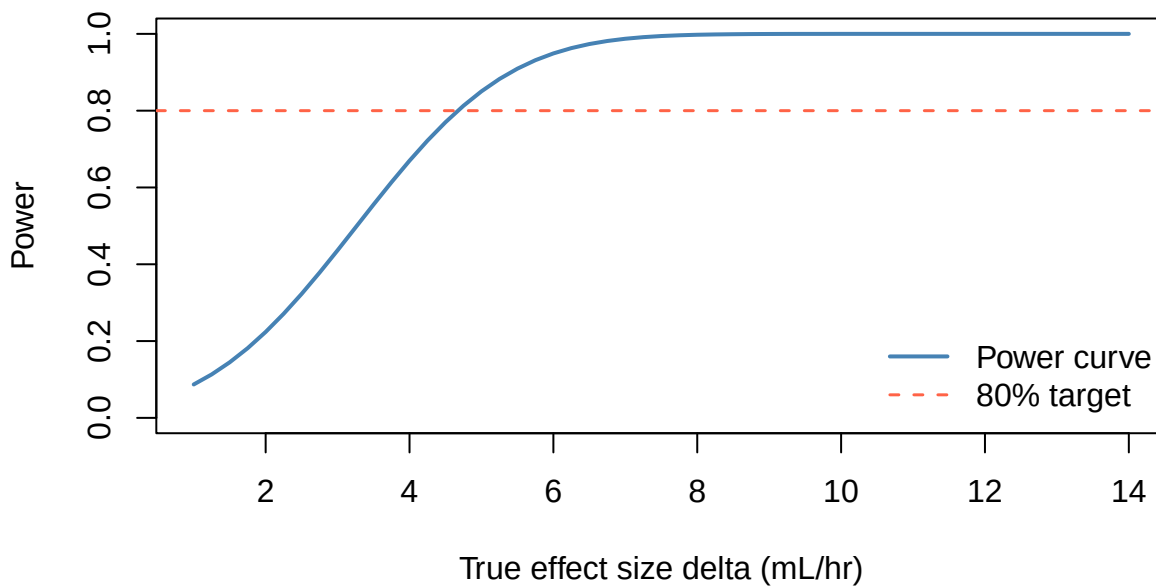
Effect of effect size on power

How does the true shift δ affect power (for $n = 25$, $\alpha = 0.05$)?

```
deltas <- seq(1, 14, by = 0.25)
pwr_delta <- sapply(deltas, function(d) {
  power.t.test(n = 25, delta = d, sd = 8,
    sig.level = 0.05, type = "one.sample",
    alternative = "two.sided")$power
})

plot(deltas, pwr_delta, type = "l", col = "steelblue", lwd = 2,
  xlab = "True effect size delta (mL/hr)", ylab = "Power",
  main = "Power vs. Effect Size (n = 25, sd = 8, alpha = 0.05)",
  ylim = c(0, 1))
abline(h = 0.80, col = "tomato", lty = 2, lwd = 1.5)
legend("bottomright",
  legend = c("Power curve", "80% target"),
  col = c("steelblue", "tomato"),
  lwd = c(2, 1.5), lty = c(1, 2),
  bty = "n")
```

Power vs. Effect Size (n = 25, sd = 8, alpha = 0.05)



Larger true effects are easier to detect — power increases with δ . Intuitively, a large signal stands out more clearly against the background noise.

Summary

What changes?	Consequence of change	Effect on power
Sample size $n \uparrow$	SE decreases	Power $\uparrow$
True effect size $\delta \uparrow$	Signal grows	Power $\uparrow$
Significance level $\alpha \uparrow$	Rejection region widens	Power $\uparrow$ (but more false positives)
Population SD $\sigma \uparrow$	SE increases, SNR drops	Power $\downarrow$